

Read-only reporting interface note

Mesa Lantern Energy LLC | Production Accounting | 2026-09-26 | v0.1-draft

This technical note describes the direction of modeled exports into Reporting Mart and the cautions for joining them. This draft is prepared for systems analysts as part of Mesa Lantern Energy's initial information handoff. It describes the modeled operation at the stated date and should be read with the period and coverage of its supporting records.

Purpose and immediate use

Field Log and Meter Historian contribute to reviewed production records, while Work Register, Inspection Register, and Vendor Ledger retain their own transactions. The reporting copy is read only. Each extract needs its source name, period, extraction timestamp, and stable business keys. A later source correction should appear as a new version in the reporting view, not as an invisible overwrite.

System roles and handoffs

Field Log and Meter Historian feed the reviewed Production Ledger. Inspection Register and Work Register exchange references when a finding leads to maintenance. Vendor Ledger carries contractor transactions. Reporting Mart receives read-only exports for monthly summaries; it does not write changes back to operational systems. Source entries and timestamped revisions remain in those systems, which is why a report should name its extraction date.

Record linkage

Synthetic well, asset, location, vendor, work-order, and project keys connect the proposed datasets. Business users should see natural asset and project names where possible, while the keys preserve joins across extracts. A linked inspection and work order should refer to the same asset instance and location. A record without a valid link should be flagged for review rather than attached to the nearest plausible item.

Monthly close boundary

Production Accounting freezes the monthly reporting snapshot on the tenth business day after month end. Changes after that point use a dated adjustment record rather than overwriting the frozen view. A reader comparing two months must therefore distinguish the original close from a later restatement view. The reporting period and extraction date are both necessary to interpret a number.

Coverage and missing values

The proposed extracts end on August 31, 2026. Production rows represent selected well-days, about 32 of 118 modeled wells per day on average, so their sums are not companywide production totals. Some meter or downtime values may be null, with a reason where available. A missing value is excluded from a numeric calculation and must never be silently treated as zero.

Null and exception treatment

A blank numeric field is not evidence of no activity. Proposed extracts can contain missing meter readings or downtime values, with a reason where one was recorded. Keep nulls visible in row review and exclude them from numeric aggregation. If an analyst needs to replace a missing value, the replacement must travel through the source owner's correction process rather than through an unexplained spreadsheet edit.

Record question for review

A join between an inspection and a work order should be rejected if the referenced equipment instance differs, even when the dates and pad happen to look plausible. The interface can carry a reference; business validation must still test whether the reference points to the intended object.

Review action and handoff

Join work orders to inspection findings only through recorded references, and verify the same asset and location. Treat missing links as exceptions. The proposed schemas describe intended joins, but generated rows must be checked before any cross-register metric is published.